

Reviewed Paper Title: *Measuring Accuracy and Consistency of Large Language Models for OOAD Principles*

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Summary

This paper evaluates whether LLMs truly understand object-oriented design, focusing on the SOLID principles. Llama 4 17b is selected as the tested agent as it is an accessible open-source model. The authors adopt a curated dataset of code snippets that either violate or follow SOLID principles and add their own annotations. They compare the model's ratings with human scores and test whether responses stay stable under small prompt changes. Results show only weak alignment with human judgment, suggesting that the model understands OO principles superficially. The consistency experiment is still in progress and may offer further insight into the model's design-level reasoning.

Comment

We like the topic of the paper and the extensive related work section. With the growing use of LLM-based coding tools, it is important to understand how well these models adhere to established coding principles. As potential next steps, the paper would benefit from deeper discussion on what kind of design-level reasoning LLMs tend to fail at. Also, adding qualitative examples alongside the existing quantitative results would help illustrate where model judgments diverge from human annotations. Finally, evaluating smaller open-source models under the same setup would offer useful insight into how model size influences their coding capabilities.

Suggestions

- The SOLID principles should be explained in the abstract for smoother reading.
- Including a code snippet with syntax highlighting in the dataset section would help clarify the examples.
- Showing a sample LLM output in the accuracy section would better illustrate how the model fails to identify proper SOLID practices.
- Adding a comparison study by doing exact experiments with smaller LLMs.

References To Consider

- *Is Your Code Generated by ChatGPT Really Correct? Rigorous Evaluation of Large Language Models for Code Generation* <https://arxiv.org/abs/2305.01210>
- *CoRe: Benchmarking LLMs Code Reasoning Capabilities through Static Analysis Tasks* <https://arxiv.org/abs/2507.05269>
- *A Survey on Code Generation with LLM-based Agents* <https://arxiv.org/abs/2508.00083v1>